

Enhanced Dev Count Tool - Headless Mode with Automated CI/CD Support

Overview

The Dev Count tool has been enhanced with powerful headless mode capabilities that enable fully automated contributor analysis in CI/CD pipelines. The tool now automatically generates reports, tracks historical data, and pushes results back to your repository.

What's New

Automated CI/CD Integration

- **Fully automated workflows:** Run contributor analysis on schedule or trigger manually
- **Automatic file generation:** Creates dated reports and CSV files automatically
- **Git push integration:** Automatically commits and pushes results back to your repository
- **Platform support:** Works with GitHub Actions, GitLab CI/CD, and Azure DevOps pipelines

Enhanced Reporting

- **Dated summary files:** `scm_summary_{YYYYMMDD}.xlsx` for historical tracking
- **Run CSV files:** Detailed contributor data per run (`run_{system}_{YYYYMMDD}.csv`)
- **Summary averages:** Tracks contributor counts across all runs with averages (`scm_summary_average.csv`)
- **Repository management:** Automatically updates and tracks repository inclusion/exclusion settings

Custom Configuration Support

- **Custom repository files:** Load repository Excel files from any location
- **Custom config files:** Use different configuration files per environment
- **Flexible paths:** All generated files respect original file locations

Quick Start Guide

1. Initial Setup (One-Time)

Run the tool in interactive mode once to set up your configuration:

```
node dist/index.js
```

Follow the prompts to configure your CI systems (GitHub, GitLab, or Azure DevOps).

2. Set Up CI/CD Pipeline

Choose your platform and follow the setup:

GitHub Actions

1. Copy `examples/veracode-dev-count-github-workflow.yml.example` to `.github/workflows/dev-count.yml`
2. Copy `examples/veracode-devcount-github.yml.example` to your repository root as `veracode-devcount-github.yml`
3. Create a GitHub secret `GITHUB_PAT` with a Personal Access Token (repo scope)

4. Update the config file with your settings
5. Commit and push - the workflow will run automatically

GitLab CI/CD

1. Copy `examples/veracode-dev-count-gitlab-workflow.yml.example` to `.gitlab-ci.yml` in your repository
2. Copy `examples/veracode-devcount-gitlab.yml.example` to your repository root as `veracode-devcount-gitlab.yml`
3. Create a GitLab CI/CD variable `GITLAB_TOKEN` with a Personal Access Token (write_repository scope)
4. Update the config file with your settings
5. Commit and push - the pipeline will run automatically

Azure DevOps

1. Copy `examples/veracode-dev-count-azuredevops-workflow.yml.example` to `azure-pipelines.yml` in your repository
2. Copy `examples/veracode-devcount-azuredevops.yml.example` to your repository root as `veracode-devcount-azuredevops.yml`
3. Create an Azure DevOps variable `AZURE_DEVOPS_TOKEN` with a Personal Access Token (Code Read & Write scope)
4. Update the config file with your organization name
5. Commit and push - the pipeline will run automatically

3. Repository File Management

The tool automatically manages `repositories-{system}.xlsx` files:

- **First run:** Creates the file with all found repositories
- **Subsequent runs:** Updates the file with new repositories while preserving your inclusion/exclusion settings
- **Custom locations:** Use `--repositories-file-{system}` to load from and update files in custom locations
- **Automatic push:** Updated repository files are pushed back to the repository

4. Viewing Results

After each run, check the `dev-count-runs/` folder in your repository:

- `scm_summary_{YYYYMMDD}.xlsx` - Detailed summary for this run
- `run_{system}_{YYYYMMDD}.csv` - Contributor details per repository
- `scm_summary_average.csv` - Historical tracking with averages
- `repositories-{system}.xlsx` - Updated repository list with inclusion settings

Key Features

Automated Contributor Tracking

- Tracks unique contributors across all repositories
- Filters contributors based on email patterns (configurable)
- Maintains separate counts for included and excluded contributors
- Generates both Excel and CSV formats for different use cases

Historical Data

- All runs are tracked in `scm_summary_average.csv`

- Easy comparison of contributor counts over time
- Automatic average calculation across all runs

Repository Management

- Excel-based repository inclusion/exclusion
- Preserves your settings across runs
- Automatically discovers new repositories
- Updates "Last Updated" timestamps

Token Requirements

GitHub

- **Token:** Personal Access Token with `repo` scope
- **Environment Variable:** `GITHUB_PAT` (preferred) or `GITHUB_TOKEN`
- **Note:** `GITHUB_TOKEN` in GitHub Actions has limited permissions - use `GITHUB_PAT` for full access

GitLab

- **Token:** Personal Access Token with `write_repository` scope
- **Environment Variable:** `GITLAB_TOKEN`
- **Note:** `CI_JOB_TOKEN` typically only has read permissions and cannot push changes

Azure DevOps

- **Token:** Personal Access Token with `Code (Read & Write)` scope
- **Environment Variable:** `AZURE_DEVOPS_TOKEN` or `SYSTEM_ACCESSTOKEN` (auto-provided in pipelines)
- **Note:** `SYSTEM_ACCESSTOKEN` is automatically used in Azure Pipelines

Example Workflow

```
# Interactive mode (one-time setup)
node dist/index.js

# Headless mode (for CI/CD)
node dist/index.js --headless \
  --mode fetch \
  --ci-systems github \
  --force-reload \
  --config veracode-devcount-github.yml \
  --repositories-file-github repositories-github.xlsx
```

Troubleshooting

Files Not Being Pushed

- **Check token permissions:** Ensure your token has write permissions
- **Verify git repository:** Must be run in a git repository
- **Check environment variables:** Ensure token is set correctly
- **Review logs:** Check for authentication errors

No Contributors Found

- **Check repository inclusion:** Verify repositories are marked "Y" in the Excel file
- **Check date range:** Commits are fetched from the last 90 days
- **Verify token permissions:** Token must have access to read commits

Permission Errors

- **GitHub:** Use `GITHUB_PAT` instead of `GITHUB_TOKEN` in CI environments
- **GitLab:** Use `GITLAB_TOKEN` (PAT) - `CI_JOB_TOKEN` cannot push
- **Azure DevOps:** Use `AZURE_DEVOPS_TOKEN` or ensure `SYSTEM_ACCESSTOKEN` has write permissions

Support

For questions or issues:

- Check the `README.md` for detailed documentation
- Review example files in the `examples/` directory
- Check workflow logs for detailed error messages

Benefits

- ✅ **Automated:** Set it and forget it - runs on schedule
- ✅ **Historical Tracking:** Track contributor trends over time
- ✅ **Centralized:** All results stored in your repository
- ✅ **Flexible:** Works with GitHub, GitLab, and Azure DevOps
- ✅ **Maintainable:** Excel-based repository management
- ✅ **Comprehensive:** Both detailed and summary reports